

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

Experience

Executive Board Member at Radyo Boğaziçi

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

Regional Organizational Team Member at ESTIEM

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

Technical Projects (can be seen at GitHub)

Performo Enterprise Admin — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

- High-density administrative intelligence dashboard with real-time operational metrics and server-side synchronized state management.

AI-Powered FinTech Intelligence Platform — Next.js, FastAPI, PostgreSQL, Docker

- Full-stack financial dashboard featuring an asynchronous limit-order matching engine and AI-generated insights.

SchedSim CPU Scheduler — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine built with zero libc dependencies using direct Linux syscalls.

Spireslayer Command Interpreter — C, Custom Backtracking Lexer

- Robust CLI-based interpreter utilizing an autonomous backtracking lexical parser for complex command resolution.

Aura Functional Interpreter — Python, Functional Paradigms, Dual Scoping Engine

- Advanced functional language interpreter supporting first-class closures and toggleable lexical/dynamic scoping engines.

MatrixNet — Java, Tarjan's SCC Algorithm, Graph Theory

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection for infrastructure vulnerability detection.

GigMatch-Pro — Java, Memory-Optimized Heaps

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for high-efficiency priority allocation.

Nightpass Survival Game — Java, Rule Engine Architecture

- Algorithmic rule engine governing strategic resource economics and state-machine logic in a terminal-based survival game.

Volunteer Experience

Education Mentor at TOÇEV

09/2019 – 03/2020

- Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds.

Volunteer at Kabataş Boys' High School Community Involvement Projects Club

09/2018 – 06/2023

- Carried out various volunteer projects aimed at helping and supporting people in need through community involvement initiatives.

Volunteering Division Member at Boğaziçi University Industrial Engineering Club

09/2023 – 06/2024

- Supported the organization and execution of volunteer initiatives and community-oriented projects within the club.

Education Mentor at Derece Kampüsü

- Mentored high school students for the university entrance exams, providing academic guidance and exam strategy support.

Technical Skills

- **Low-Level & Systems:** x86-64 Assembly, C, Linux Syscalls, OS Internals (Scheduling, Pthreads, Memory Mgmt)
- **Enterprise Full-Stack:** Next.js 15+, TypeScript/React, FastAPI, ORM (Prisma, SQLAlchemy), PostgreSQL/SQLite, Tailwind & Framer Motion
- **Algorithms & Theory:** Graph Theory (Tarjan's SCC), Advanced Data Structures (Heaps), Complexity Analysis (Big O), Recursive Optimization
- **Paradigms & Models:** Functional (Scheme), Logic (Prolog), OOP (Java/C++), Interpreter Design, Finite State Machines
- **AI & Agentic Systems:** Advanced Agentic Coding, AI-Assisted Architecture, LLM Engineering (OpenAI API, Prompting)
- **Engineering Ecosystem:** Linux/Unix, Git/GitHub, Docker, CI/CD, Alembic, Pytest, Makefile, LaTeX

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually